

Product Line Profitability & Margin Performance Analysis

Nassau Candy Distributor — Detailed Research Paper

Abstract

This study evaluates 10,194 order-line records to determine which products and divisions create economic value, where margins are weak, and how concentrated the portfolio is. Total sales are **\$141,783.63**, gross profit is **\$93,442.80**, and weighted gross margin is **65.91%**. Chocolate is the dominant profit engine, contributing 95.06% of gross profit. The analysis identifies Kazookles as the clearest margin-risk product at approximately 7.69% gross margin and the Other division as the weakest major division by margin. A dashboard translates the results into interactive product, division, cost and concentration diagnostics.

1. Business Context

For a distributor, sales volume alone can be misleading. A high-sales product can generate limited profit when manufacturing cost is high, while a smaller product can have strong unit economics. The project therefore evaluates gross margin, profit per unit, revenue contribution and profit contribution at product and division level.

2. Objectives

The objectives are to rank products by gross profit and margin, compare division-level revenue and profit, identify high-sales/low-margin risk, quantify Pareto concentration, diagnose cost-versus-sales relationships, measure margin stability, and provide recommendations for repricing, sourcing and portfolio review.

3. Dataset

The dataset contains 10,194 order-line records, with 8,549 distinct orders and 15 distinct products. Order dates cover 2024-01-02 through 2025-12-12. Fields include customer and geography information, division, product, sales, units, gross profit and cost.

4. Data Cleaning and Validation

Date fields were converted to datetime and financial fields to numeric values. Gross margin was calculated as $\text{Gross Profit} / \text{Sales} \times 100$ and profit per unit as $\text{Gross Profit} / \text{Units}$. Gross profit reconciles to Sales minus Cost in the supplied financial fields. A major anomaly was found in shipment dates: the supplied ship dates extend into 2026–2030 while orders are dated 2024–2025, producing extremely long intervals. Shipment timing is therefore excluded from business conclusions until validated.

5. KPI Framework

KPI	Formula	Interpretation
Gross Margin	Gross Profit / Sales	Profitability of revenue
Profit per Unit	Gross Profit / Units	Unit economics
Revenue Contribution	Product Sales / Total Sales	Revenue dependency
Profit Contribution	Product Profit / Total Profit	Profit dependency
Margin Volatility	Std. dev. monthly margin	Stability over time

Company-wide weighted gross margin is **65.91%**. Monthly gross-margin standard deviation is approximately **1.11 percentage points**, indicating relatively stable aggregate margin performance.

6. Product-Level Profitability

Product aggregation reveals an extremely concentrated profit structure. The five leading products are all Chocolate Wonka Bar products and together generate approximately 95.1% of total gross profit. The leading product is Wonka Bar - Scrumdiddlyumptious, followed by Triple Dazzle Caramel, Milk Chocolate, Nutty Crunch Surprise and Fudge Mallows.

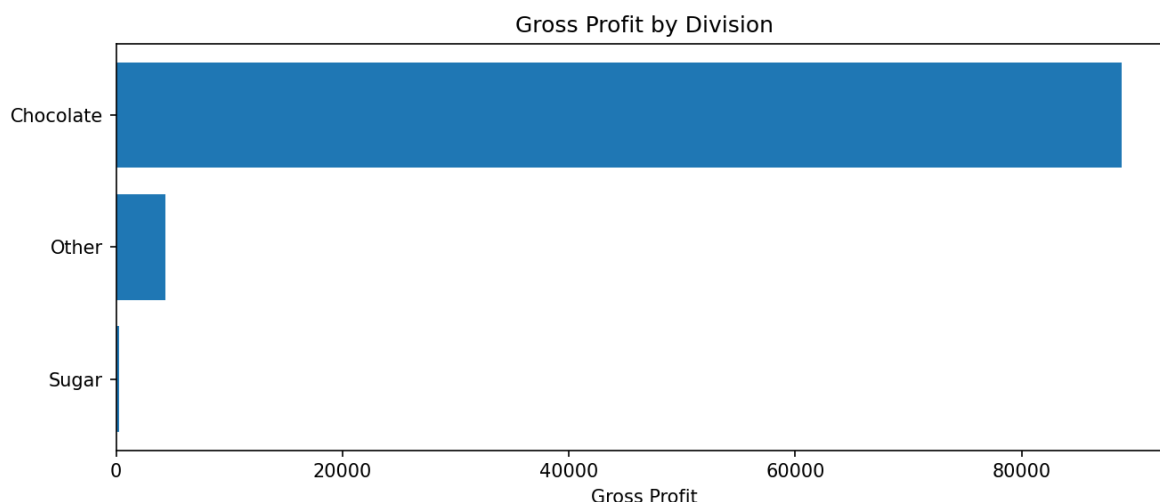
Product	Sales	Gross Profit	Margin	Profit %
Wonka Bar -Scrumdiddlyumptious	\$27,874.80	\$19,357.50	69.44%	20.72%
Wonka Bar - Triple Dazzle Cara	\$28,485.00	\$18,610.20	65.33%	19.92%
Wonka Bar - Milk Chocolate	\$26,867.75	\$17,443.37	64.92%	18.67%
Wonka Bar - Nutty Crunch Surpr	\$23,574.95	\$16,819.95	71.35%	18.00%
Wonka Bar - Fudge Mallows	\$24,890.40	\$16,593.60	66.67%	17.76%

The top four products alone contribute approximately **77.30%** of total gross profit; the top five contribute **95.06%**. This creates dependency risk but also identifies a clear priority group for price, cost and availability monitoring.

7. Division Performance

Division analysis shows a strong revenue-to-profit imbalance in the Other division. Chocolate generates 92.88% of revenue and 95.06% of gross profit at a 67.45% margin. Other generates 6.82% of revenue but only 4.64% of profit at 44.84% margin. Sugar has a healthy 66.61% margin but represents only 0.30% of revenue and profit.

Division	Sales	Profit	Margin	Revenue %	Profit %
Chocolate	\$131,692.90	\$88,824.62	67.45%	92.88%	95.06%
Other	\$9,663.25	\$4,333.45	44.84%	6.82%	4.64%
Sugar	\$427.48	\$284.73	66.61%	0.30%	0.30%



8. Pareto Profit Concentration

The portfolio has very high profit concentration. Approximately 77.30% of profit comes from the top four products and 95.06% from the top five. Management should protect these products while monitoring whether lower-contribution products justify their operational complexity.

9. Margin Risk

The lowest-margin products are:

Product	Division	Sales	Profit	Margin
Kazookles	Other	\$1,205.75	\$92.75	7.69%
Fun Dip	Sugar	\$12.00	\$4.80	40.00%
Nerds	Sugar	\$15.00	\$7.00	46.67%

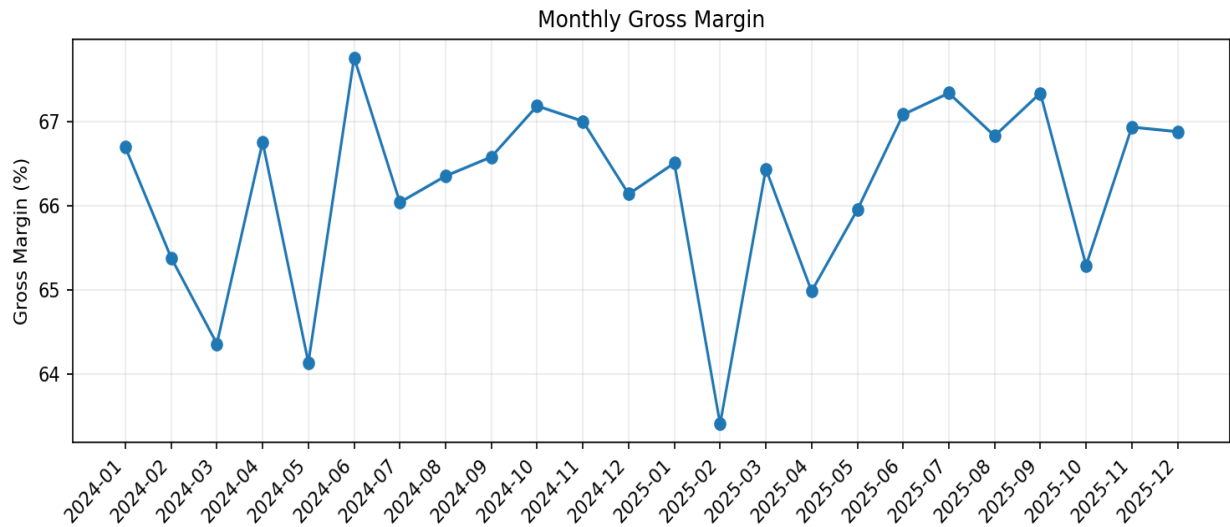
Product	Division	Sales	Profit	Margin
SweeTARTS	Sugar	\$61.50	\$28.70	46.67%
Lickable Wallpaper	Other	\$7,860.00	\$3,930.00	50.00%

Kazookles is the clearest margin-risk case, with approximately 7.69% gross margin. Other sub-50% products include Fun Dip, Nerds and SweeTARTS. These products should be reviewed for price realization, sourcing cost, discounts and strategic importance before any discontinuation decision.

10. Cost Structure Diagnostics

Cost-versus-sales analysis is used to identify products where cost consumes a disproportionate share of revenue. The diagnostic supports targeted review rather than automatic action. Supplier cost, customer pricing, promotions and volume economics should be examined for flagged products.

11. Margin Stability



Monthly gross margin volatility is approximately 1.11 percentage points. This suggests the aggregate margin profile is relatively stable over time, while individual products can still have very different unit economics.

12. Key Findings

- Overall weighted gross margin is 65.91%.
- Chocolate contributes 95.06% of gross profit and is the core profit engine.
- Five products contribute about 95.1% of total gross profit.
- The Other division has materially weaker margin at 44.84%.
- Kazookles has the lowest margin at approximately 7.69%.
- Aggregate monthly margin is relatively stable.
- Shipment-date data is anomalous and requires validation.

13. Recommendations

1. Protect core Chocolate products: monitor pricing, sourcing cost and availability. **2. Review Kazookles:** investigate repricing and cost reduction before rationalization. **3. Investigate Other:** identify sourcing, pricing or assortment drivers of its weaker margin. **4. Use profit contribution:** evaluate products on profit as well as sales. **5. Define official margin thresholds:** the dashboard threshold is an analytical screen, not corporate policy. **6. Monitor concentration:** track top-product dependence over time. **7. Validate ship dates:** correct the source before using shipment timing for logistics decisions.

14. Dashboard Design

The Streamlit application provides date-range selection, division filtering, product search and a margin-threshold slider. It presents headline sales/profit/margin KPIs, product-level profitability, division comparisons, cost-versus-sales diagnostics, margin-risk flags and profit concentration.

15. Limitations

Only 15 distinct products are represented, so concentration is naturally high. The dataset does not provide supplier prices, promotions, discounts, inventory, official margin targets or competitor pricing. Therefore, the analysis identifies decision priorities but does not establish causal drivers. The shipment-date anomaly limits logistics interpretation.

16. Conclusion

The portfolio produces a strong 65.91% overall gross margin, but the profit pool is highly concentrated in Chocolate. The Other division and low-margin products such as Kazookles are the clearest areas for management review. The dashboard converts these findings into an actionable framework for product rationalization, repricing and sourcing discussions.